

GONÇALO FRUTUOSO

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Summary

Software Engineer and MSc Student specializing in **Distributed Systems** and **Computational Biology**. Currently combining industry experience at **Sky** (Apple Ecosystem) with high-performance computing research at **INESC-ID**. Focused on building scalable architectures, optimizing complex algorithms, and delivering high-quality production code.

Education

Instituto Superior Técnico - Universidade de Lisboa **Sep 2024 – Present**
Mestrado em Engenharia Informática e de Computadores — **Current Grade: 16/20** *Lisboa, Portugal*

- Double Majoring in **Distributed Systems** and **Computational Biology**
- Master's Thesis: "**On large-scale trees and networks exploratory analysis and visualization**" (Ongoing)
- Also taking courses in other majors that align with my interests

Instituto Superior de Engenharia de Lisboa **Sep 2021 – Jul 2024**
Licenciatura em Engenharia Informática e de Computadores — **Final Grade: 14/20** *Lisboa, Portugal*

- Strong skills in Computer and Software Engineering
- Thesis: **SCAR: Smart Solution Academic Registry** — Grade: **18/20**
- **180.5 ECTS credits** — EQF Level 6

Technical Skills

Languages: Kotlin, Java, Python, JavaScript, TypeScript, Solidity, Swift, Groovy

Backend Development: Spring, Node.js, Nginx, Docker, Jenkins

Frontend Development: React.js, React Native, Jetpack Compose, SwiftUI, HTML, CSS, Bootstrap, Webpack

Databases: PostgreSQL, MongoDB, Elasticsearch, Firestore, MySQL

Tools & Platforms: Git, GitHub, GCP, AWS, Linux, LaTeX, Jira and Confluence

Other: Android, Blockchain Development

Work Experience

Sky Portugal **Jan 2026 – Present**
Associate Software Developer *Lisbon, Portugal*

- Contributing to the core development of the **Apple TV** application using **Swift**, serving millions of users globally.
- Collaborating in an **Agile** environment to implement new features and maintain high availability of the streaming platform.
- Participating in architectural discussions and code reviews to ensure scalability and adherence to Clean Code principles.

INESC-ID Lisboa **Nov 2025 – Present**
Research Assistant *Lisbon, Portugal*

- Conducting research in **Phylogenetics** and **High-Performance Computing (HPC)** for large-scale graph visualization.
- Investigating efficient algorithms to optimize data processing pipelines for complex network structures.

Sky Portugal **Jul 2025 – Aug 2025**
Frontend Apprentice *Lisbon, Portugal*

- Developed features for the **Apple TV** ecosystem using **Swift**, directly impacting the user interface experience.
- Implemented **CI/CD pipelines** using **Jenkins** and **Groovy**, optimizing the build and deployment workflow.
- Integrated into a cross-functional team, adopting DevOps culture and Scrum methodologies.

Universidade Lusófona **Jun 2023 – Oct 2023**
Programmer *Lisbon, Portugal*

- Developed a data acquisition GUI using **Python** and **CustomTkinter** for a Tension Testing Machine.
- Improved data accuracy and workflow efficiency through user-centric interface design.

Projects

SCAR: Smart Contract Academic Registry

Blockchain — Solidity — React Native

- Developed **DiGo Certify App**, a blockchain-based platform to verify academic credentials and combat fraud
- Implemented **Solidity smart contracts** ensuring GDPR-compliant, tamper-proof credential validation
- Integrated Web3 functionality with Expo React Native for a seamless user interface

Games@Cloud

Java — AWS (EC2, DynamoDB) — Machine Learning — Distributed Systems

- Architected an intelligent load balancer with **ML-based workload prediction** and **3 adaptive scheduling strategies** (Spreading, Packing, Hybrid) inspired by OpenStack Nova, managing computational workloads across AWS EC2 instances
- Implemented **dynamic auto-scaling system** and bytecode instrumentation framework using Javassist to transparently collect runtime metrics and automatically manage EC2 instance lifecycle
- Engineered complete infrastructure automation with shell scripts managing IAM roles, security groups, AMI creation, and deployment to AWS

Gomoku Full-Stack Game

Kotlin — Spring Boot — React — TypeScript — Android

- Built backend REST APIs in Kotlin and Spring Boot, with ReactJS frontend and native Android app for unified gaming experience
- Deployed using Nginx reverse proxy for secure and efficient distribution

DepChain: Permissioned Blockchain System

Java — Distributed Systems — Consensus Protocols

- Architected a permissioned blockchain system implementing a custom fault-tolerant consensus algorithm
- Focused on modular client-server architecture with reconfiguration and dependability mechanisms

ChargeIST: Electric Vehicle Charging Management App

Android — Kotlin — Spring Boot — PostgreSQL

- Developed a mobile app integrating Google Maps SDK for real-time EV charging station discovery
- Implemented a Spring Boot backend with authentication, WebSocket updates, and containerized PostgreSQL
- Followed **MVVM architecture** using Jetpack Compose, Kotlin Coroutines, Hilt DI, and Ktor HTTP client

Awards & Achievements

Deloitte Merit Award

Apr 2025

- Awarded for achieving the **best group performance** in the Software Architecture course at Instituto Superior Técnico

Interests & Hobbies

Sports & Outdoor Activities

Ongoing

- **Boxing:** Developed balance between personal and academic life
- **Skimboarding & Surfing:** Passionate about water sports and ocean activities
- **Motocross:** 4-wheel motorcycle riding since age 7, honing risk management and precision

Technical Hobbies

Ongoing

- **Car Restoration:** Hands-on mechanical work and problem-solving through automotive projects